

STANDARD OPERATING PROCEDURE

Instructional Brief: Comprehensive Bank Statement Analysis

1. Document Ingestion & Verification

The objective of this phase is to transform unstructured financial documents into a machine-readable format while ensuring data integrity.

Protocol:

- **Source Identification:** Identify if the statement is a native digital PDF, a scanned image, or a CSV export.
- **OCR Processing:** For non-digital PDFs, utilize high-fidelity OCR tools. Ensure that the row-column alignment is preserved to prevent data "shifting."
- **Validation:** Cross-reference the *Opening Balance* and *Closing Balance* listed on the statement header with the sum of all transactions.

2. Data Normalization & Structuring

All extracted data must be mapped to a standardized schema. This allows for cross-account comparisons and automated financial modeling.

Field Name	Technical Requirement	Format/Type
Transaction Date	The date the transaction was finalized.	YYYY-MM-DD
Description	The raw merchant string provided by the bank.	String
Transaction Category	Primary spending bucket (e.g., Housing, Travel).	Categorical
Debit (-)	Outbound funds/Expenses.	Decimal/Currency

Field Name	Technical Requirement	Format/Type
Credit (+)	Inbound funds/Income.	Decimal/Currency
Running Balance	The balance remaining after the transaction.	Decimal/Currency

3. Key Detail Extraction & Analysis

Once structured, the data must be cleaned and audited for specific financial insights.

3.1 Merchant Normalization

Remove extraneous metadata from the description field (e.g., "PURCHASE AUTHORIZED ON 05/20") to isolate the core vendor name.

3.2 Categorization Logic

- **Fixed Costs:** Flag recurring monthly amounts (e.g., Rent, Utilities).
- **Discretionary:** Categorize lifestyle spending (e.g., Dining, Entertainment).
- **Financial:** Identify transfers, bank fees, and interest payments.

3.3 Anomaly Detection

Search for and flag the following:

- ☐ Duplicate transactions within a 24-hour window.
- ☐ Subscription spikes (unexpected increases in recurring amounts).
- ☐ Large "one-off" debits exceeding 20% of the average monthly spend.

Final Quality Gate: The analysis is considered complete only when the Total Credits - Total Debits exactly matches the Net Change in the account balance for the period.